

Fisher's Separation Theorem: Analytical Example

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1: Consumption with Perfect Capital Market

Optimal Consumption plan

- Consider a two-period world where a single consumer must decide between consumption today (c_o) and consumption tomorrow (c_1).
- The consumer's initial endowments are m_o today and m_1 tomorrow respectively.
- The consumer has the opportunity to borrow or lend b_o amount today at interest rate (r).
- The equations governing the consumer's feasible actions today and tomorrow are:

$$c_o = m_o + b_o \tag{1}$$

$$c_1 = m_1 - (1 + r)b_o \tag{2}$$

- Since consumption in both periods must be non-negative, the limit on how much the consumer may borrow or lend today is given by $-m_o \leq b_o \leq \frac{m_1}{1+r}$.

- The consumer's *budget constraint* is given by (substituting $c_0 - m_o$ for b_o in (2))

$$c_o + \frac{c_1}{1+r} = m_o + \frac{m_1}{1+r} := W_o. \quad (3)$$

- The budget constraint is written as an equality since we shall assume our consumer always prefers more consumption.
- We can interpret W_o as the consumer's *present value of wealth*.
- Consequently, the *future value of wealth* is $(1+r)W_o$.

- Suppose that the consumer has a utility function

$$U(c_o, c_1) = \sqrt{c_o} + \beta\sqrt{c_1}$$

- The parameter β serves as a *discount factor or subjective rate of time preference* on future consumption. Smaller values imply a larger discount factor on future consumption, which implies that our consumer prefers more consumption today.
- Formally, the consumer's optimization problem is given by:

$$Max [U(c_o, c_1) : c_o + \frac{c_1}{1+r} = W_o], \quad (4)$$

which may be equivalently expressed as

$$Max [U(c_o, c_1(c_o)) : 0 \leq c_o \leq W_o] \quad (5)$$

where

$$c_1(c_o) = (1+r)(W_o - c_o) \quad (6)$$

- The form of utility implies that consumption in both periods must be positive so that the optimal choice for c_o lies strictly between 0 and W_o .
- The first-order optimality condition imply,

$$0 = \frac{\partial U}{\partial c_o} + \frac{\partial U}{\partial c_1} \frac{dc_1}{dc_o} = \frac{1}{2\sqrt{c_o}} - \frac{\beta(1+r)}{2\sqrt{c_1}}. \quad (7)$$

- From (7) the optimal consumption plan necessarily satisfies the condition

$$\sqrt{\frac{c_1}{c_o}} = \beta(1+r). \quad (8)$$

or equivalently

$$c_1 = (\beta)^2(1+r)^2 c_o \quad (9)$$

- Substituting (9) into the budget constraint (3) and solving for c_o and c_1 , the optimal consumption plan is given by

$$c_o = \left[\frac{1}{1 + (\beta)^2(1 + r)} \right] W_o = \rho_1 W_o \quad (10)$$

$$c_1 = \left[\frac{((\beta)^2)((1 + r)^2)}{1 + (\beta)^2(1 + r)} \right] W_o = \rho_1 W_o \quad (11)$$

- It is important to note that the constants ρ_o and ρ_1 are independent of the present wealth W_o , that is, they are known parameters strictly determined by the two discount factors β and r respectively.
- The optimal utility is of the form

$$U^*(W_o) = [\sqrt{\rho_o} + \beta\sqrt{\rho_1}] \sqrt{W_o} \quad (12)$$

Numerical example: 1

- Let $m_o = 100, m_1 = 990, r = 0.10, \beta = 0.6$.
- Then calculate $c_0, c_1, U^*(W_o)$.
- Here $W_o = 1000$ [**why?**]
- Therefore, what is the value of $c_o, c_1, U^*(W_o)$?
- Has the utility increased or decreased due to the capital market?
- If increased, by how much?

2: Consumption and Investment with Perfect Capital Market

- In the same model, now let us assume that the consumer has the opportunity to invest I_o today in production from which he will receive $f(I_o)$ tomorrow.
- The function $f(\cdot)$ encapsulates the investment opportunities in production available to our consumer.
- $f(\cdot)$ is a well behaved function with the partial derivatives of 'correct' signs - including $f'(0) = +\infty$ (Inada conditions)
- The equations governing the consumer's feasible actions today and tomorrow are:

$$c_o + I_o = m_o + b_o \quad (13)$$

$$c_1 = m_1 + f(I_o) - (1 + r)b_o. \quad (14)$$

- The new *budget constraint* is

$$c_o + \frac{c_1}{1+r} = W_o + \left[\frac{f(I_o)}{1+r} - I_o \right] = W_o \quad (15)$$

- Eq. (15) reveals a fundamental property: All consumers, regardless of their utility function, should first determine the optimal investment plan to increase their wealth.
- That is, they should select the value of I_o such that the marginal return $f'(I_o) = (1+r)$.

- In general, for any investment opportunities $f(I_o)$ of a firm owned by shareholders (consumers), the pareto optimal solution is for the firm to optimize its investment *regardless* of the individual shareholders' subjective time preference (or different desires) for consumption today versus tomorrow.
- Because there exists a frictionless capital market, each consumer can redistribute between today and tomorrow to increase their wealth as they desire. This principle is known as the *Fisher Separation Theorem*.

Numerical example: 2

- Suppose $f(I_o) = (33\sqrt{I_o})$. What is the optimal choice of I_o ?
- The additional wealth created through investment equals - ?
- So what is the value of $\hat{W}_o$?
- What is the optimal consumption plan in this case?
- Has the utility increased or decreased? If increased, by how much?
- To obtain the optimal consumption plan, the consumer must borrow $b_0 = ?$ today, pay back ? tomorrow, thereby leaving him/her with ? to consume in the final period.

2: Consumption and Investment Without a Capital Market

- Let us consider the situation in which the consumer has investment opportunities as described in the previous section, but no longer has the opportunity to borrow or lend, i.e. $b_o = 0$.
- Show that without access to a capital market our consumer is far worse off.
- The equations governing the consumer's feasible actions today and tomorrow are now:

$$c_o + I_o = m_o \quad (16)$$

$$c_1 = m_1 + f(I_o). \quad (17)$$

- The new budget constraint is represented as:

$$c_1(c_o) = m_1 + f(m_o - c_o). \quad (18)$$

- The first-order optimality condition imply

$$\sqrt{\frac{c_1}{c_o}} = \beta f'(I_o), \quad (19)$$

or, equivalently, the optimal choice for I_o must satisfy

$$\sqrt{\frac{m_1 + f(I_o)}{m_o - I_o}} = \beta f'(I_o). \quad (20)$$

- Substitute the specific choice for f and find the optimal choice for I_o must satisfy ?

$$990 + 33\sqrt{I_o} = \frac{9801}{I_o} - 98.01 \quad (21)$$

- The optimal value for $I_o = 8.25$ and the corresponding $c_o = 91.75, c_1 = 1085, U^* = 29.34$. The utility has dropped considerably to the level corresponding to the original endowment.